AI-Augmented Human Biocuration in Reactome

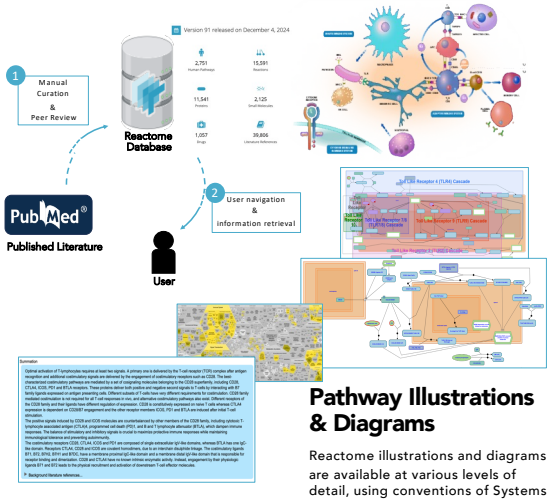
Marc Gillespie[1,4], Nancy T Li[1], Deidre Beavers[2], Helia Mohammadi[1, 9], Adam Wright[1], Marija Milacic[1], Patrick Conley[2], Chuqiao Gong[3], Johannes Griss[3,5], Robin Haw[1], Bijay Jassal[1], Lisa Matthews[6], Bruce May[1], Eliot Ragueneau[3], Karen Rothfels[1], Cristoffer Sevilla[3], Veronica Shamovsky[6], Ralf Stephan[1,7], Krishna Tiwari [3], Thawfeek Varusai[3,8], Joel Weiser[1], Henning Hermjakob[3, *] Peter D'Eustachio[6, *], Guanming Wu [2, *], Lincoln Stein[1,9,*]

[1] Ontario Institute for Cancer Research, Toronto, Canada, [2]Oregon Health & Science University, Portland,USA, [3] European Molecular Biology Laboratory, European Bioinformatics Institute (EMBL-EBI), Hinxton, UK, [4] College of Pharmacy and Health Sciences, St. John's University, Queens, USA, [5] Department of Dermatology, Medical University of Vienna, Austria, [6] NYU Grossman School of Medicine, New York, USA, [7] Institute for Globally Distributed Open Research and Education (IGDORE), [8] Open Targets, Wellcome Genome Campus, [9]Department of Molecular Genetics, University of Toronto

The Reactome Curation Process

The Reactome Knowledgebase is a public resource renowned for its meticulous human curation and peer-review practices[1].

As the most comprehensive open-source, human biological pathway knowledgebase, Reactome has operated for 23 years with its expert curation driven by staff scientists, resulting in 2,751 human pathways.



Pathway Summations

Pathways in Reactome are organized hierarchically, with summations at each level that describe the pathways, subpathways, or events.

Pathway summations are designed for human-readability, and contain references to primary literature, experimental evidence, and review articles for further reading.

Pathway Illustrations & Diagrams

Reactome illustrations and diagrams are available at various levels of detail, using conventions of Systems Biology Graphical Notations (SBGN).

The illustrations can be overlaid with user-provided data, providing an interactive visual framework for data representation and analysis.

LLM-augmentation of the Curation Process

Reactome's curation process faces inherent challenges in handling the vastness and complexity of biological data.

To address this, we are developing a web-based curation tool that incorporates Large Language Models (LLM) technology, guided by Reactome's data schema and curation requirements, to address key bottlenecks for the curation workflow, while also remaining human-driven.

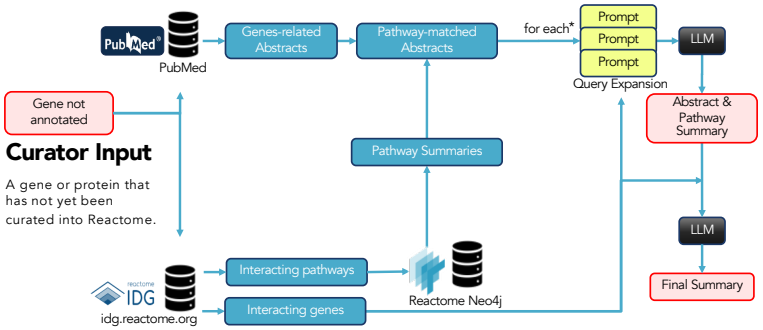
RAG-grounded with PubMed

We adopted the RAG (retrieval-augmented generation) technology, which provides the LLM with reliable text references, such as PubMed.

By grounding LLM responses in reliable references, RAG minimizes the risk of hallucinations compared to naïve LLM.

LLM Integration

Reactome summaries and Pubmed abstracts are provided to the LLM to extract relevant evidence and functional relationships.



Curator Input

A gene or protein that has not yet been curated into Reactome.

Integration with IDG[2]

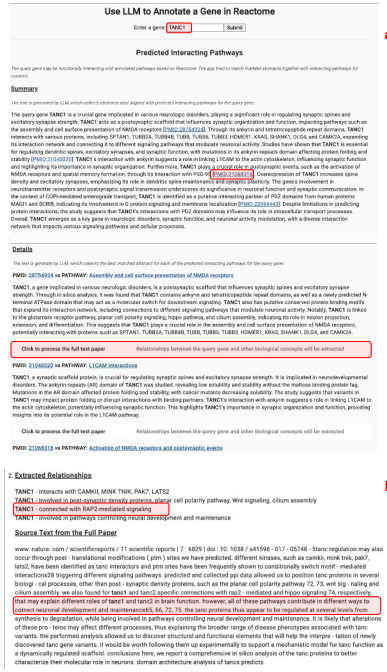
We associate previously unannotated genes with Reactome pathways using our prior work with the NIH IDG (Illuminating the Dark Genome) program, predicting possible functional relationships from literature.

LLM Output

Text summaries, functional relationships, and supportive evidence are provided to the curator, forming a skeleton for curated pathway reports.

Results

The new curator tool will allow curators to search for genes, as shown below, during their curation process.



TANC1 has not been annotated in Reactome yet.

Summary for query gene generated by LLM contains links to literature

Relevant details are collected and summarized for identified literature

Predicted pathways are identified

Relevant source text is displayed

Conclusions

Our LLM-focused approaches lay the groundwork for more scalable and efficient curation workflows within Reactome, with potential applications for community-driven contributions.

reactome.org

References:

1 Milacic et al, 2024, *Nucleic Acids Research*. "The reactome pathway knowledgebase 2024" DOI: 10.3389/fimmu.2023.1282859
2 Beavers et al, 2023. *Curr Protocols*. Illuminate the Functions of Dark Proteins Using the Reactome-IDG Web Portal" DOI: 10.1002/cpz1.845.

*Supported by NIH grants. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.